NS-EN 1992-1-1:2004+A1:2014+NA:2024

2.3.3 (3)	5.8.3.3 (1)	6.8.7 (1)	9.8.3 (1)
2.4.2.1 (1)	5.8.3.3 (2)	7.2 (2)	9.8.3 (2)
2.4.2.2 (1)	5.8.5 (1)	7.2 (3)	9.8.4 (1)
2.4.2.2 (2)	5.8.6 (3)	7.2 (5)	9.8.5 (3)
2.4.2.2 (3)	5.10.1 (6)	7.3.1 (5)	9.10.2.2 (2)
2.4.2.3 (1)	5.10.2.1 (1)P	7.3.2 (4)	9.10.2.3 (3)
2.4.2.4 (1)	5.10.2.1 (2)	7.3.4 (3)	9.10.2.3 (4)
2.4.2.4 (2)	5.10.2.2 (4)	7.4.2 (2)	9.10.2.4 (2)
2.4.2.5 (2)	5.10.2.2 (5)	8.2 (2)	11.3.5 (1)P
3.1.2 (2)P	5.10.3 (2)	8.3 (2)	11.3.5 (2)P
3.1.2 (4)	5.10.8 (2)	8.6 (2)	11.3.7 (1)
3.1.6 (1)P	5.10.8 (3)	8.8 (1)	11.6.1 (1)
3.1.6 (2)P	5.10.9 (1)P	9.2.1.1 (1)	11.6.1 (2)
3.2.2 (3)P	6.2.2 (1)	9.2.1.1 (3)	11.6.2 (1)
3.2.7 (2)	6.2.2 (6)	9.2.1.2 (1)	11.6.4.1 (1)
3.3.4 (5)	6.2.3 (2)	9.2.1.4 (1)	12.3.1 (1)
3.3.6 (7)	6.2.3 (3)	9.2.2 (4)	12.6.3 (2)
4.4.1.2 (3)	6.2.4 (4)	9.2.2 (5)	A.2.1 (1)
4.4.1.2 (5)	6.2.4 (6)	9.2.2 (6)	A.2.1 (2)
4.4.1.2 (6)	6.4.3 (6)	9.2.2 (7)	A.2.2 (1)
4.4.1.2 (7)	6.4.4 (1)	9.2.2 (8)	A.2.2 (2)
4.4.1.2 (8)	6.4.5 (1)	9.3.1.1(3)	A.2.3 (1)
4.4.1.2 (13)	6.4.5 (3)	9.5.2 (1)	C.1 (1)
4.4.1.3 (1)P	6.4.5 (4)	9.5.2 (2)	C.1 (3)
4.4.1.3 (3)	6.5.2 (2)	9.5.2 (3)	E.1 (2)
4.4.1.3 (4)	6.5.4 (4)	9.5.3 (3)	J.1 (2)
5.1.3 (1)P	6.5.4 (6)	9.6.2 (1)	J.2.2 (2)
5.2 (5)	6.8.4 (1)	9.6.3 (1)	J.3 (2)
5.5 (4)	6.8.4 (5)	9.7 (1)	J.3 (3)
5.6.3 (4)	6.8.6 (1)	9.8.1 (3)	
5.8.3.1 (1)	6.8.6 (3)	9.8.2.1 (1)	

Table 3.1 Strength and deformation characteristics for concrete

	Strength classes for concrete														Analytical relation / Explanation
f_{ck} (MPa)	12	16	20	25	30	35	40	45	50	55	60	70	80	90	
$f_{ck,cube}$ (MPa)	15	20	25	30	37	45	50	55	60	67	75	85	95	105	
f_{cm} (MPa)	20	24	28	33	38	43	48	53	58	63	68	78	88	98	$f_{cm} = f_{ck}+8$(MPa)
f_{ctm} (MPa)	1,6	1,9	2,2	2,6	2,9	3,2	3,5	3,8	4,1	4,2	4,4	4,6	4,8	5,0	$f_{ctm}=0{,}30\times f_{ck}^{(2/3)} \leq$C50/60 $f_{ctm}=2{,}12\cdot\ln(1+(f_{cm}/10))$ > C50/60
$f_{ctk,\,0,05}$ (MPa)	1,1	1,3	1,5	1,8	2,0	2,2	2,5	2,7	2,9	3,0	3,1	3,2	3,4	3,5	$f_{ctk;0,05} = 0{,}7\times f_{ctm}$ 5% fractile
$f_{ctk,0,95}$ (MPa)	2,0	2,5	2,9	3,3	3,8	4,2	4,6	4,9	5,3	5,5	5,7	6,0	6,3	6,6	$f_{ctk;0,95} = 1{,}3\times f_{ctm}$ 95% fractile
E_{cm} (GPa)	27	29	30	31	33	34	35	36	37	38	39	41	42	44	$E_{cm} = 22[(f_{cm})/10]^{0,3}$ (f_{cm} in MPa)
ε_{c1} (‰)	1,8	1,9	2,0	2,1	2,2	2,25	2,3	2,4	2,45	2,5	2,6	2,7	2,8	2,8	see Figure 3.2 $\varepsilon_{c1}\,(^0/_{00}) = 0{,}7\,f_{cm}^{\;0,31} < 2.8$
ε_{cu1} (‰)	3,5									3,2	3,0	2,8	2,8	2,8	see Figure 3.2 for $f_{ck} \geq 50$ Mpa $\varepsilon_{cu1}(^0/_{00})=2{,}8+27[(98-f_{cm})/100]^4$
ε_{c2} (‰)	2,0									2,2	2,3	2,4	2, 5	2,6	see Figure 3.3 for $f_{ck} \geq 50$ Mpa $\varepsilon_{c2}(^0/_{00})=2{,}0+0{,}085(f_{ck}-50)^{0,53}$
ε_{cu2} (‰)	3,5									3,1	2,9	2,7	2,6	2,6	see Figure 3.3 for $f_{ck} \geq 50$ Mpa $\varepsilon_{cu2}(^0/_{00})=2{,}6+35[(90-f_{ck})/100]^4$
n	2,0									1,75	1,6	1,45	1,4	1,4	for $f_{ck} \geq 50$ Mpa $n=1{,}4+23{,}4[(90-f_{ck})/100]^4$
ε_{c3} (‰)	1,75									1,8	1,9	2,0	2,2	2,3	see Figure 3.4 for $f_{ck} \geq 50$ Mpa $\varepsilon_{c3}(^0/_{00})=1{,}75+0{,}55[(f_{ck}-50)/40]$
ε_{cu3} (‰)	3,5									3,1	2,9	2,7	2,6	2,6	see Figure 3.4 for $f_{ck} \geq 50$ Mpa $\varepsilon_{cu3}(^0/_{00})=2{,}6+35[(90-f_{ck})/100]^4$

2.3.3 (3)
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3.1.6 (2)P
3.2.2 (3)P
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4.4.1.3 (1)P
4.4.1.3 (3)
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5.8.6 (3)
5.10.1 (6)
5.10.2.1 (1)P
5.10.2.1 (2)
5.10.2.2 (4)
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5.10.9 (1)P
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9.10.2.4 (2)
11.3.5 (1)P
11.3.5 (2)P
11.3.7 (1)
11.6.1 (1)
11.6.1 (2)
11.6.2 (1)
11.6.4.1 (1)
12.3.1 (1)
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A.2.1 (1)
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A.2.2 (1)
A.2.2 (2)
A.2.3 (1)
C.1 (1)
C.1 (3)
E.1 (2)
J.1 (3)
J.2.2 (2)
J.3 (2)
J.3 (3)

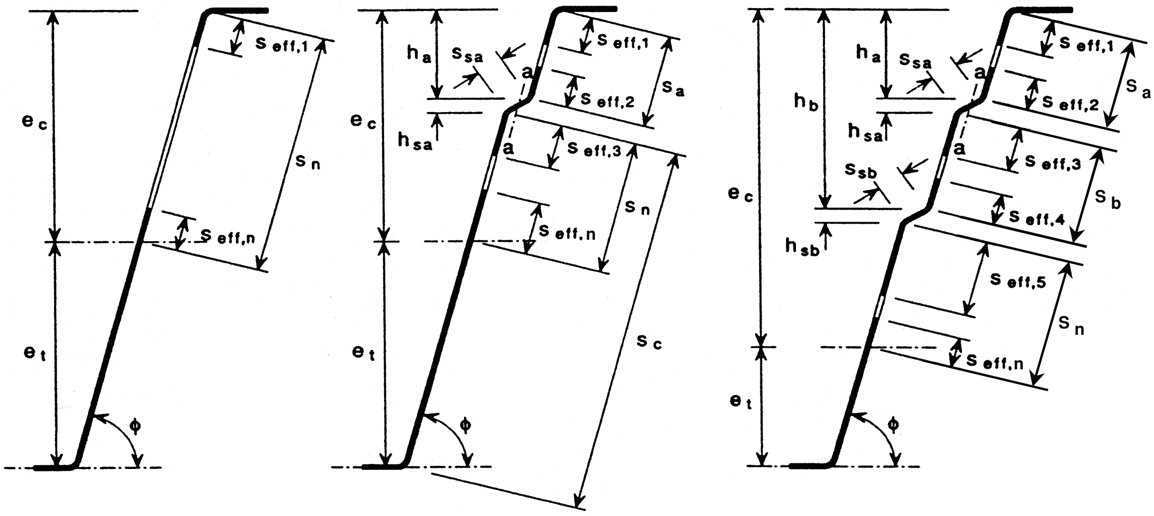


Figure 5.12: Effective cross-sections of webs of trapezoidal profiled sheets

(3) The effective areas of the stiffeners should be obtained from the following:

- for a single stiffener, or for the stiffener closer to the compression flange:

$$A_{sa} = t\,(s_{eff,2} + s_{eff,3} + s_{sa}) \qquad \text{... (5.30)}$$

- for a second stiffener:

$$A_{sb} = t\,(s_{eff,4} + s_{eff,5} + s_{sb}) \qquad \text{... (5.31)}$$

in which the dimensions $s_{eff,1}$ to $s_{eff,n}$ and s_{sa} and s_{sb} are as shown in figure 5.12.

(4) Initially the location of the effective centroidal axis should be based on the effective cross-sections of the flanges but the gross cross-sections of the webs. In this case the basic effective width $s_{eff,0}$ should be obtained from:

$$s_{eff,0} = 0{,}76\,t\,\sqrt{E\,/\left(\gamma_{M0}\,\sigma_{com,Ed}\right)} \qquad \text{... (5.32)}$$

where:

$\sigma_{com,Ed}$ is the stress in the compression flange when the cross-section resistance is reached.

(5) If the web is not fully effective, the dimensions $s_{eff,1}$ to $s_{eff,n}$ should be determined as follows:

$$s_{eff,1} = s_{eff,0} \qquad \text{... (5.33a)}$$

$$s_{eff,2} = (1 + 0{,}5 h_a / e_c)\, s_{eff,0} \qquad \text{... (5.33b)}$$

$$s_{eff,3} = [1 + 0{,}5(h_a + h_{sa}) / e_c]\, s_{eff,0} \qquad \text{... (5.33c)}$$

$$s_{eff,4} = (1 + 0{,}5 h_b / e_c)\, s_{eff,0} \qquad \text{... (5.33d)}$$

$$s_{eff,5} = [1 + 0{,}5(h_b + h_{sb}) / e_c]\, s_{eff,0} \qquad \text{... (5.33e)}$$

$$s_{eff,n} = 1{,}5 s_{eff,0} \qquad \text{... (5.33f)}$$

where:

e_c is the distance from the effective centroidal axis to the system line of the compression flange, see figure 5.12;

and the dimensions h_a, h_b, h_{sa} and h_{sb} are as shown in figure 5.12.

(6) The dimensions $s_{\mathrm{eff},1}$ to $s_{\mathrm{eff},n}$ should initially be determined from (5) and then revised if the relevant plane element is fully effective, using the following:

- in an unstiffened web, if $s_{\mathrm{eff},1} + s_{\mathrm{eff},n} \geq s_n$ the entire web is effective, so revise as follows:

$$s_{\mathrm{eff},1} = 0{,}4 s_n \qquad \text{... (5.34a)}$$

$$s_{\mathrm{eff},n} = 0{,}6 s_n \qquad \text{... (5.34b)}$$

- in stiffened web, if $s_{\mathrm{eff},1} + s_{\mathrm{eff},2} \geq s_a$ the whole of s_a is effective, so revise as follows:

$$s_{\mathrm{eff},1} = \frac{s_a}{2 + 0{,}5 h_a / e_c} \qquad \text{... (5.35a)}$$

$$s_{\mathrm{eff},2} = s_a \frac{(1 + 0{,}5 h_a / e_c)}{2 + 0{,}5 h_a / e_c} \qquad \text{... (5.35b)}$$

- in a web with one stiffener, if $s_{\mathrm{eff},3} + s_{\mathrm{eff},n} \geq s_n$ the whole of s_n is effective, so revise as follows:

$$s_{\mathrm{eff},3} = s_n \frac{[1 + 0{,}5(h_a + h_{sa})/e_c]}{2{,}5 + 0{,}5(h_a + h_{sa})/e_c} \qquad \text{... (5.36a)}$$

$$s_{\mathrm{eff},n} = \frac{1{,}5 s_n}{2{,}5 + 0{,}5(h_a + h_{sa})/e_c} \qquad \text{... (5.36b)}$$

- in a web with two stiffeners:

  - if $s_{\mathrm{eff},3} + s_{\mathrm{eff},4} \geq s_b$ the whole of s_b is effective, so revise as follows:

$$s_{\mathrm{eff},3} = s_b \frac{1 + 0{,}5(h_a + h_{sa})/e_c}{2 + 0{,}5(h_a + h_{sa} + h_b)/e_c} \qquad \text{... (5.37a)}$$

$$s_{\mathrm{eff},4} = s_b \frac{1 + 0{,}5 h_b / e_c}{2 + 0{,}5(h_a + h_{sa} + h_b)/e_c} \qquad \text{... (5.37b)}$$

  - if $s_{\mathrm{eff},5} + s_{\mathrm{eff},n} \geq s_n$ the whole of s_n is effective, so revise as follows:

$$s_{\mathrm{eff},5} = s_n \frac{1 + 0{,}5(h_b + h_{sb})/e_c}{2{,}5 + 0{,}5(h_b + h_{sb})/e_c} \qquad \text{... (5.38a)}$$

$$s_{\mathrm{eff},n} = \frac{1{,}5 s_n}{2{,}5 + 0{,}5(h_b + h_{sb})/e_c} \qquad \text{... (5.38b)}$$

(2) For the design of spread foundations, appropriately simplified models for the description of the soil-structure interaction may be used.

Note: For simple pad footings and pile caps the effects of soil-structure interaction may usually be ignored.

(3) For the strength design of individual piles the actions should be determined taking into account the interaction between the piles, the pile cap and the supporting soil.

(4) Where the piles are located in several rows, the action on each pile should be evaluated by considering the interaction between the piles.

(5) This interaction may be ignored when the clear distance between the piles is greater than two times the pile diameter.

5.1.3 Load cases and combinations

(1)P In considering the combinations of actions, see EN 1990 Section 6, the relevant cases shall be considered to enable the critical design conditions to be established at all sections, within the structure or part of the structure considered.

Note: Where a simplification in the number of load arrangements for use in a Country is required, reference is made to its National Annex. The following simplified load arrangements are recommended for buildings:

(a) alternate spans carrying the design variable and permanent load ($\gamma_Q Q_k + \gamma_G G_k + P_m$), other spans carrying only the design permanent load, $\gamma_G G_k + P_m$ and

(b) any two adjacent spans carrying the design variable and permanent loads ($\gamma_Q Q_k + \gamma_G G_k + P_m$). All other spans carrying only the design permanent load, $\gamma_G G_k + P_m$.

5.1.4 Second order effects

(1)P Second order effects (see EN 1990 Section 1) shall be taken into account where they are likely to affect the overall stability of a structure significantly and for the attainment of the ultimate limit state at critical sections.

(2) Second order effects should be taken into account according to 5.8.

(3) For buildings, second order effects below certain limits may be ignored (see 5.8.2 (6)).

5.2 Geometric imperfections

(1)P The unfavourable effects of possible deviations in the geometry of the structure and the position of loads shall be taken into account in the analysis of members and structures.

Note: Deviations in cross section dimensions are normally taken into account in the material safety factors. These should not be included in structural analysis. A minimum eccentricity for cross section design is given in 6.1 (4).

(2)P Imperfections shall be taken into account in ultimate limit states in persistent and accidental design situations.

(3) Imperfections need not be considered for serviceability limit states.

(4) The following provisions apply for members with axial compression and structures with vertical load, mainly in buildings. Numerical values are related to normal execution deviations

(Class 1 in ENV 13670). With the use of other deviations (e.g. Class 2), values should be adjusted accordingly.

(5) Imperfections may be represented by an inclination, θ_i, given by:

$$\theta_i = \theta_0 \cdot \alpha_h \cdot \alpha_m \tag{5.1}$$

where
- θ_0 is the basic value:
- α_h is the reduction factor for length or height: $\alpha_h = 2/\sqrt{l}$; $2/3 \le \alpha_h \le 1$
- α_m is the reduction factor for number of members: $\alpha_m = \sqrt{0,5(1+1/m)}$
- l is the length or height [m], see (4)
- m is the number of vertical members contributing to the total effect

Note: The value of θ_0 for use in a Country may be found in its National Annex. The recommended value is 1/200

(6) In Expression (5.1), the definition of l and m depends on the effect considered, for which three main cases can be distinguished (see also Figure 5.1):

- Effect on isolated member: l = actual length of member, m =1.
- Effect on bracing system: l = height of building, m = number of vertical members contributing to the horizontal force on the bracing system.
- Effect on floor or roof diaphragms distributing the horizontal loads: l = storey height, m = number of vertical elements in the storey(s) contributing to the total horizontal force on the floor.

(7) For isolated members (see 5.8.1), the effect of imperfections may be taken into account in two alternative ways a) or b):

a) as an eccentricity, e_i, given by

$$e_i = \theta_i \, l_0 / 2 \tag{5.2}$$

where l_0 is the effective length, see 5.8.3.2

For walls and isolated columns in braced systems, $e_i = l_0/400$ may always be used as a simplification, corresponding to $\alpha_h = 1$.

b) as a transverse force, H_i, in the position that gives maximum moment:

for unbraced members (see Figure 5.1 a1):

$$H_i = \theta_i N \tag{5.3a}$$

for braced members (see Figure 5.1 a2):

$$H_i = 2\theta_i N \tag{5.3b}$$

where N is the axial load

Note: Eccentricity is suitable for statically determinate members, whereas transverse load can be used for both determinate and indeterminate members. The force H_i may be substituted by some other equivalent transverse action.

NS-EN 1992-1-1:2004+A1:2014+NA:2024

$$\nu = 0,6(1 - f_{ck}/250) \quad \text{(NA.6.6N)}$$

NA.6.2.3 Members requiring design shear reinforcement

NA.6.2.3(2) The value of cotθ may be chosen in the range determined by formula (NA.6.7.aN), except in sections with significant axial tension ($\sigma_t = N_{Ed}/A_c \geq f_{ctk,\ 0,05}$). There cot$\theta$ is determined by the formula (NA.6.7.bN).

$$1,0 \leq \cot\theta \leq 2,5 \quad \text{(NA.6.7.aN)}$$

$$1,0 \leq \cot\theta \leq 1,25 \ (\text{if } \sigma_t \geq f_{ctk,\ 0,05}) \quad \text{(NA.6.7.bN)}$$

NA.6.2.3(3) The factor ν_1 may be set equal to ν determined by formula NA.6.6N. If the tension in the shear reinforcement is less than 0,8$f_{yk,}\nu_1$ may be determined on the basis of formulas (NA.6.10.aN) and (NA.6.10.bN).

$$\nu_1 = 0,6 \text{ for } f_{ck} \leq 60 \text{ MPa} \quad \text{(NA.6.10.aN)}$$

$$\nu_1 = 0,9 - f_{ck}/200 > 0,5 \text{ for } f_{ck} > 60 \text{ MPa} \quad \text{(NA.6.10.bN)}$$

The factor α_{cw} is set equal to 1 in structures without prestressing or axial compression. In structures with prestressing or axial compression, α_{cw} is determined on the basis of the formulas (NA.6.11.aN), (NA.6.11.bN) and (NA.6.11.cN).

$$\alpha_{cw} = (1 + \sigma_{cp}/f_{cd}) \text{ for } 0 < \sigma_{cp} \leq 0,25\ f_{cd} \quad \text{(NA.6.11.aN)}$$

$$\alpha_{cw} = 1,25 \text{ for } 0,25\ f_{cd} < \sigma_{cp} \leq 0,5\ f_{cd} \quad \text{(NA.6.11.bN)}$$

$$\alpha_{cw} = 2,5(1 - \sigma_{cp}/f_{cd}) \text{ for } 0,5\ f_{cd} < \sigma_{cp} < 1,0\ f_{cd} \quad \text{(NA.6.11.cN)}$$

where σ_{cp} is the mean compressive stress, which is specified in greater detail in note 3 of the standard.

NA.6.2.4 Shear between web and flanges of T-sections

NA.6.2.4(4) The value of cotθ may be chosen in the range determined by formulas (NA.6.7.aN) and (NA.6.7.bN):

$$1,0 \leq \cot\theta \leq 2,5 \text{ in flanges with compression} \quad \text{(NA.6.7.aN)}$$

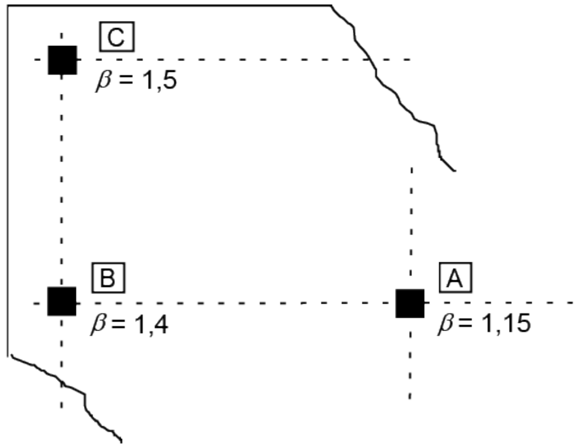
$$1,0 \leq \cot\theta \leq 1,25 \text{ in flanges with tension} \quad \text{(NA.6.7.bN)}$$

NA.6.2.4(6) The factor k is set equal to 0,4.

NA.6.4.3 Punching shear calculation

NA.6.4.3(6) Simplified values of the factor β are given in Figure NA.6.21N.

NS-EN 1992-1-1:2004+A1:2014+NA:2024



A internal column

B edge column

C corner column

Figure NA.6.21N — Simplified values for β

NA.6.4.4 Punching shear resistance of slabs and column bases without shear reinforcement

NA.6.4.4(1) The factor $C_{Rd,c}$ is set equal to k_2/γ_C

$$C_{Rd,c} = k_2/\gamma_C$$

where

k_2 = 0,15 if the conditions for using k_2 equals 0,18 are not met

k_2 = 0,18 for concrete with the maximum size of aggregate D, according to NS-EN 12620, greater than or equal to 16 mm, and where the coarse aggregate constitutes (approximately) 50% or more of the total content of aggregate and where coarse aggregates with a low strength (fragmentation coefficient $LA > LA_{40}$ for concrete strength above B35, otherwise $LA > LA_{50}$) are not used.

The factor k_1 is set equal to 0,1 for compression and equal to 0,3 for tension (σ_{cp} negative for tension).

The minimum shear resistance associated with principal tensile failure, ν_{min}, is determined as:

$$\nu_{min} = 0,035k^{3/2}f_{ck}{}^{1/2} \quad \text{(NA.6.3N)}$$

It is assumed that f_{ck} is not given a value greater than 65 MPa in formulas (6.47) and (6.3N).

NA.6.4.5 Punching shear resistance of slabs and column bases with shear reinforcement

NA.6.4.5(1) The value of k_{max} is set equal to 1,5 for stirrups and 1,8 for bolts/rods with T-heads or reinforcement ladders with documented capacity as shear reinforcement, stirrups and bolts/rods with T-heads are anchored around the tension reinforcement, see 9.2.2(2). The value of k_{max} is also used for light weight aggregate concrete in formula (11.6.52)